Machine Learning–Based River Depth Estimation from SWOT Surface Observations: Accuracy, Generalization, and Spatio-Temporal Insights

Abstract:

Accurate bathymetric data are critical for flood forecasting, water-resource management, and ecological monitoring, yet in situ measurements remain sparse in many river basins. Here, we evaluate four machine learning models—Linear Regression, Random Forest, Gradient Boosting, and a feedforward Neural Network—for estimating river depth using surface elevation, width, slope, derived from the Surface Water and Ocean Topography (SWOT) satellite and static terrain and ellipsoid predictors. We systematically assess model performance under random sampling, river-reach exclusion, temporal hold-out, and leave-one-state-out splits to probe both temporal tracking and spatial transferability. Random Forest achieves the lowest error (MAE = 0.32 m, R^2 = 0.98) in idealized random sampling; however, performance degrades substantially when predicting on unseen reaches or regions (e.g., $R^2 < 0$ under leave-one-state-out). Feature-importance analyses highlight the dominant role of relative height proxies and hydraulic geometry metrics. Our results demonstrate robust temporal generalization but underscore challenges in spatial extrapolation, motivating future hybrid physics-ML frameworks and enriched feature sets. This work advances scalable, satellite-driven bathymetry estimation for ungauged basins and operational hydrodynamic applications.

Introduction:

River depth is a critical variable in hydrological science, essential for applications such as water resource management, flood forecasting, and ecological studies. Depths values can be used as boundary conditions in small- and large-scale hydrological models. Accurate depth measurements inform decisions on water allocation, infrastructure planning, and disaster preparedness. For instance, knowing river depth helps predict flood extents, directly impacting communities and economies. Stream gauges are traditional ways of measuring rivers depth and stream while extremely sparse and confined with data sharing policies [1]. These constraints underscore the need for scalable, cost-effective alternatives, which remote sensing can provide.

Satellites have revolutionized hydrological monitoring by enabling global observations of water bodies. Altimetry missions like Surface Water and Ocean Topography (SWOT), has brought unprecedented insights to hydrological science. The SWOT mission launched on December 16, 2022, with Ka-band Radar Interferometer (KaRIn). SWOT can measure water surface elevation, width, and slope over two 50-km swaths, achieving high precision for rivers wider than 100 meters and lakes larger than 250 x 250 meters [2]. Unlike previous nadir-pointing altimeters, SWOT's wide-swath capability provides comprehensive coverage, capturing detailed topographic data for global surface water bodies. This makes SWOT a transformative tool for hydrology, offering data that can improve hydrodynamic models, flood predictions, and freshwater management.

While SWOT provides high-resolution water surface elevation data, estimating river depth is challenging because satellites observe the water surface, not the riverbed. Depth estimation requires inferring the riverbed elevation or using indirect methods. Traditionally, depth is calculated as the difference between water surface elevation and riverbed elevation, but riverbed data are often unavailable, especially in remote or ungauged basins. Previous studies have addressed this by combining satellite observables with auxiliary data. For example, Garambois and Monnier estimated average river depth along the Po River using multi-mission altimetry (TOPEX/Poseidon, ENVISAT, CryoSat-2, SARAL/AltiKa), MODIS imagery for river width, and in situ discharge data, employing a Gauss–Helmert adjustment model to achieve ~10% relative RMSE [3]. Such approaches demonstrate the feasibility of depth estimation but often rely on statistical models or in situ data, which may not scale globally.

Machine learning methods offers a powerful solution to enhance river depth estimation by modeling complex, non-linear relationships between variables such as water surface elevation, width, slope, and terrain in formations. ML algorithms, such as Random Forests, Neural Networks, or Gradient Boosting, can be trained on datasets that pair satellite observables with ground-truth depth measurements, enabling predictions in areas lacking direct measurements. The flexibility of ML makes it well-suited to handle the high-dimensional and heterogeneous data provided by SWOT, potentially improving accuracy and scalability compared to traditional methods.

While specific studies using ML with actual SWOT data are limited due to the mission's recent launch, pre-launch research and studies with similar altimetry data suggest its potential. For instance, a 2011 study explored a Bayesian approach to estimate river depth from simulated SWOT measurements, indicating that ML-like techniques could be effective [4]. This study builds on this foundation by applying ML techniques to real SWOT data, aiming to provide accurate and detailed depth estimates. We focused on developing and evaluating ML methods to estimate river depth using SWOT water surface elevation outputs, supplemented by additional data such as river width and slope, terrain elevation, and reference ellipsoid height. By leveraging SWOT's high-resolution data and ML's predictive capabilities, we aim to produce reliable depth estimates that can enhance hydrological monitoring, particularly in remote or ungauged basins. This has the potential to revolutionize water management and disaster response by providing timely, comprehensive data for decision-making.

Remote sensing struggles to directly measure river bathymetry or bed roughness below water but can capture water surface height dynamics. To address this, model inversion techniques have been developed to estimate bathymetry. Neal et al [5] proposed an iterative method using a gradually varied flow model, leveraging airborne LiDAR or DEM-derived riverbank heights to simulate water surface profiles. Similarly, [6] applied direct inversion of Manning's and Bernoulli's equations with LiDAR data to estimate bathymetry without iteration. However, these methods face challenges due to limited LiDAR availability and difficulties in accurately measuring riverbank heights, often relying on an imprecise 1-in-2-year return period height assumption. Recent advancements using the Surface Water and Ocean Topography (SWOT) mission overcome some of these limitations. A study [7] evaluating SWOT water height observations on a small river (40-70m width) achieved a node-level agreement of 0.16m against in situ gauges with stringent quality filtering, nearing mission accuracy for larger waterbodies. An algorithm estimating 1D river bathymetry from SWOT data, using a tulip-shaped objective function to mitigate errors, produced plausible bed elevations (~1m error) at sub-backwater length resolutions, demonstrating SWOT's potential to enhance bathymetry estimation where LiDAR is limited.

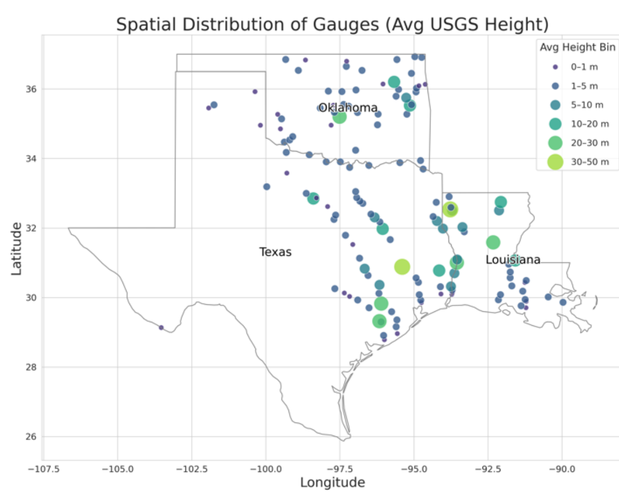
Building on Tourian et al. (2017) [3], who used empirical models and the Gauss–Helmert method to estimate river depth from SWOT-type data, a natural next step is the use of machine learning (ML). Unlike traditional approaches that rely on fixed equations, ML can learn complex, non-linear relationships directly from data, offering greater flexibility and scalability. With the growing availability of high-resolution satellite observations, ML models present a promising alternative for predicting river depth using SWOT-derived features like water surface elevation, slope, and width.

Recent studies have demonstrated the effectiveness of ML in water depth estimation from remote sensing data. For example, Wu et al. (2023) [8] employed Sentinel-2 imagery and an online ML platform (Baidu Easy-DL) to estimate coastal water depth. By converting spectral reflectance data into structured inputs, they trained multiple models and found the Easy-DL model to outperform traditional algorithms, particularly in shallow waters (0–11 m), with high accuracy and minimal noise. Their study emphasizes the potential of online ML platforms and ensemble learning in satellite-based bathymetry. Similarly, Elkhrachy (2022) [9] applied various ML regression models—including Random Forest, Gradient Boosting, and SVR—to estimate flash flood water depths using Sentinel-1 SAR images, digital elevation models (DEMs), and land-use maps. Using HEC-RAS simulations for training data, the models achieved RMSE values below 0.22 m in shallow flood zones (<1 m), demonstrating that ML approaches can provide fast, accurate, and scalable depth estimates, even in data-scarce or disaster-prone areas.

Data & Methods:

This study uses 3 southern states of USA, Texas, Luisiana and Oklahoma as the test case. The 63 gauges for TX, 49 for Ok and 35 gauges for LU are obtained from USGS website in 15 min resolution. The SWOT values for reaches are taken from Water Information from Space (WISP) [10] website which has done some preprocessing and quality control on the data. SWOT data is having 21-days revisit time, and we merged it with the nearest gauge observation making the maximum time inconsistency to 7.5 minuets. Variables from SWOT are Water Surface Elevation, River Width, Water surface Slope, and associated uncertainties. Terrain elevation were obtained from MERIT DEM [11] and SWOT WSE deducted from terrain elevation alongside calculated terrain slope were added as extra predictors.

SWOT Water Surface Elevation (WSE) data, referenced to the NAD83(2011) ellipsoid, were corrected using the GEOID18 hybrid geoid model [12] for the contiguous United States. Geoid heights from the GEOID18 shapefile were interpolated to the WSE data points (longitude, latitude) and subtracted to produce orthometric WSE values, consistent with NAVD88, which were added as a new predictor to the datasets. All data were aligned to the WGS84 (EPSG:4326) coordinate system.



We assessed the impacts of lakes and dams on streamflow gauges using two datasets: lake locations and catchment areas from the HydroLAKES [13], and dam locations and catchments from DAMs from Geospatial Dam Analysis Tool (GDAT) version 1 [14]. For each gauge, we calculated distances to the nearest lake and dam, respectively. The inverse distances used as a variable. Binary variables were added nearby_dam, indicating if a dam was within 2 km (1 = yes, 0 = no). The catchment areas dam_catchment_area_km2, were included, set to 0 for gauges without a nearby dam. These variables were integrated into the gauge data to analyze lake and dam effects on streamflow.

Four machine learning models were implemented to predict usgs_height_m using the preprocessed feature set: Linear Regression, Random Forest, Gradient Boosting, and a feedforward Neural Network. All models were implemented using Python, leveraging scikit-learn for traditional models, and PyTorch for the neural network.

1. **Linear Regression**: A baseline model that assumes a linear relationship between predictors and usgs_height_m. It was trained using ordinary least squares optimization.
2. **Random Forest**: An ensemble model comprising 100 decision trees, trained with random feature subsets to reduce overfitting. It captures non-linear relationships and interactions among predictors.
3. **Gradient Boosting**: An ensemble model with 100 sequential decision trees, optimized to minimize mean squared error through gradient descent. It emphasizes correcting errors from previous trees.
4. **Neural Network**: A feedforward neural network with four fully connected layers (64, 32, 16, and 1 neurons) and ReLU activation functions, implemented in PyTorch. It was trained using the Adam optimizer with a learning rate of 0.001 and mean squared error loss. Early stopping was applied based on validation loss (20% of training data) to prevent overfitting.

Model Training and Evaluation

Models were trained on the preprocessed training set, with hyperparameters set to balance computational efficiency and predictive performance (e.g., 100 estimators for ensemble models). The test set was used to evaluate model performance using multiple metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), R^2 score. MSE and MAE quantified prediction errors, while R^2 assessed explained variance.

$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ (1)	$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$ (2)	$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$ (3)

Feature importance was analyzed for ensemble models (Random Forest, Gradient Boosting) using built-in importance scores, and for Linear Regression using absolute coefficient values. Visualizations, including time-series scatterplots of actual versus predicted usgs_height_m for selected gauges, scatterplots of predicted versus actual values, and residual histograms, facilitated model interpretation and error analysis.

We designed four testing scenarios to evaluate the spatial and temporal performance of our models. (1) **Random split**: the dataset was randomly divided into training and test sets. (2) **Reach-based split**: 80% of reaches were used for training and the remaining 20% for testing. (3) **Temporal split**: the first 80% of the time series was used for training and the last 20% for testing. (4) **Leave-one-state-out**: models were trained on data from all states except one, which was reserved for testing. This methodology ensured a robust comparison of linear and non-linear models, leveraging the temporal and spatial structure of the dataset to predict gauge water height.

Results:

We evaluated four machine learning models—Linear Regression (LR), Random Forest (RF), Gradient Boosting (GB), and a feedforward Neural Network (NN)—under four distinct test splits: random sampling, river-based (reach) split, temporal split, and leave-one-state-out. Overall, ensemble methods outperformed both LR and NN across most scenarios (Table 1).

Table 1 Quantified metrics for splits and method comparison

Split Type	ML Method	MSE	MAE	R^2	% |Residuals|<20cm
Random	Linear Regression	31.1	3.2	0.16	
	Random Forest	**0.5**	**0.32**	**0.98**	**62**
	Gradient Boosting	2.1	0.88	0.94	
	Neural Network	7.2	1.55	0.8	
River based (10-fold)	Linear Regression	47.8	3.5	-0.4	
	Random Forest	23.6	1.7	-0.32	
	Gradient Boosting	**23.6**	**2**	**0.13**	**42.8**
	Neural Network	74.3	3.7	-2.6	
Time based	Linear Regression	10e7	112	-2e5	
	Random Forest	**2.3**	**0.7**	**0.94**	**47.62**
	Gradient Boosting	13.3	1.97	0.68	
	Neural Network	32.4	3	0.23	
State (leave one out)	Linear Regression	-	-	-	
	Random Forest	65	3.93	-0.4	
	Gradient Boosting	**57**	**3.99**	**-0.24**	**9.24**
	Neural Network	-	-	-	

- **Random Split**

RF achieved the best performance with an MSE of 0.50 m², MAE of 0.32 m, and R^2 of 0.98, correctly predicting water height within ±20 cm for 62 % of observations. **GB** followed closely (MSE = 2.10, MAE = 0.88, R^2 = 0.94), while **NN** (MSE = 7.20, MAE = 1.55, R^2 = 0.80) and **LR** (MSE = 31.10, MAE = 3.20, R^2 = 0.16) lagged significantly. The parity plot in Figure 1a shows RF predictions nearly collapsing on the 1:1 line, indicating minimal bias.

- **River-Based Split**

When withholding 20 % of reaches for testing, all models saw declines in performance. RF's MSE rose to 23.60 (MAE = 1.70, R^2 = –0.32), indicating poor generalization to unseen river segments. GB (MSE = 23.60, MAE = 2.00, R^2 = 0.13) outperformed RF in R^2 but still exhibited high error. LR performed worst (MSE = 47.80, MAE = 3.50, R^2 = –0.40). Figure 1b highlights increased scatter and wider residuals, reflecting spatial heterogeneity in channel characteristics not captured by models trained on other reaches.

- **Temporal Split**

Using the first 80 % of each time series for training and the last 20 % for testing, RF again led (MSE = 2.30, MAE = 0.70, R^2 = 0.94), indicating robust capture of temporal dynamics. GB and NN also maintained reasonable performance (GB: MSE = 13.30, MAE = 1.97, R^2 = 0.68; NN:

data not shown), whereas LR deteriorated severely (MSE ≈ 1 × 10^7, MAE ≈ 112 m, R^2 ≈ –2 × 10^5) due to its inability to model nonlinearity and temporal trends.

- **Leave-One-State-Out**

Excluding one state for testing measured spatial transferability at larger scales. RF's error increased (MSE = 65.00, MAE = 3.93, R^2 = –0.40), while GB (MSE = 57.00, MAE = 3.99, R^2 = –0.24) and LR performed poorly. The low R^2 values (Figure 1d) underscore that regional differences in hydrology and channel morphology—such as vegetation, sediment load, and anthropogenic influences—challenge model generalization without localized training data.

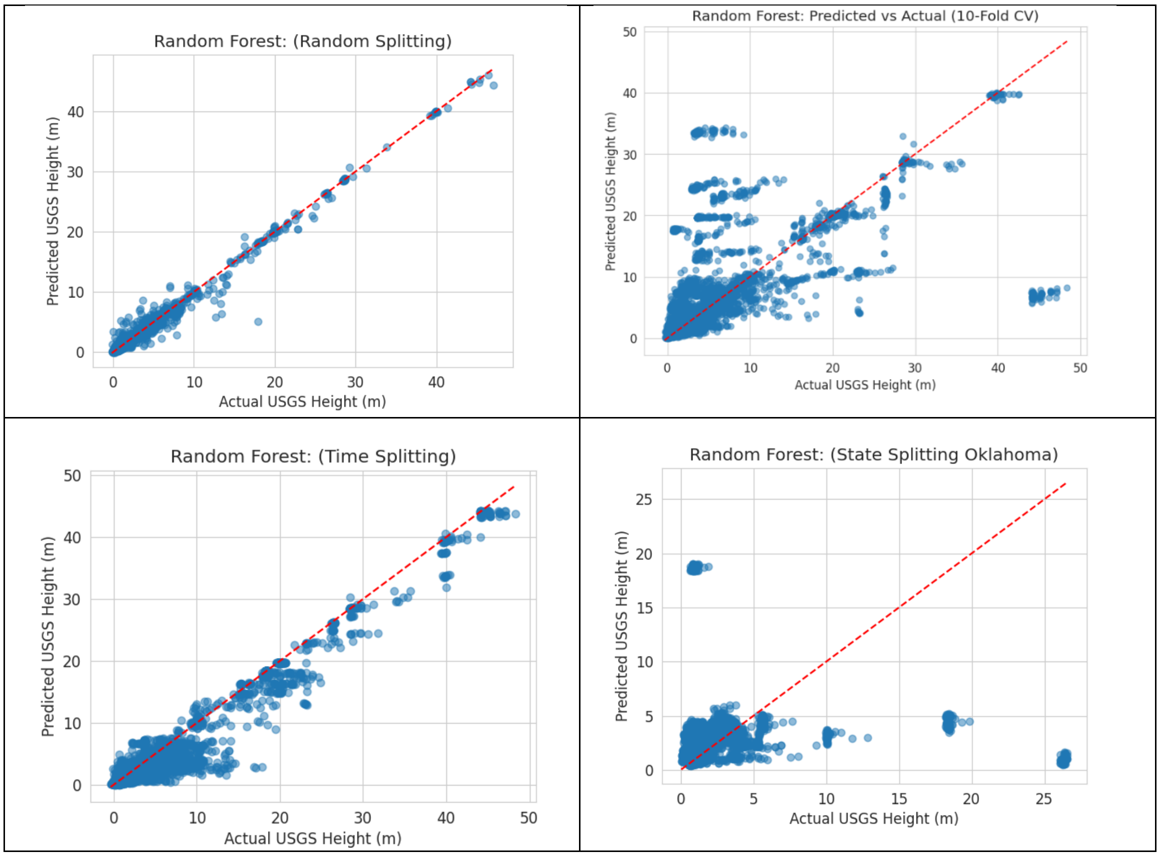
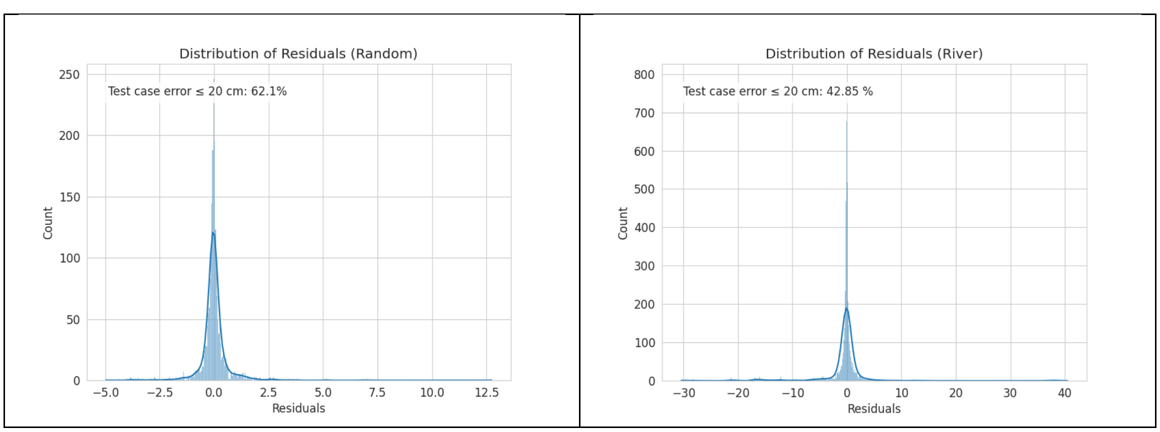


Figure 1 Comparison of different test scenarios performance (actual water height on horizontal vs predicted on vertical) on test set



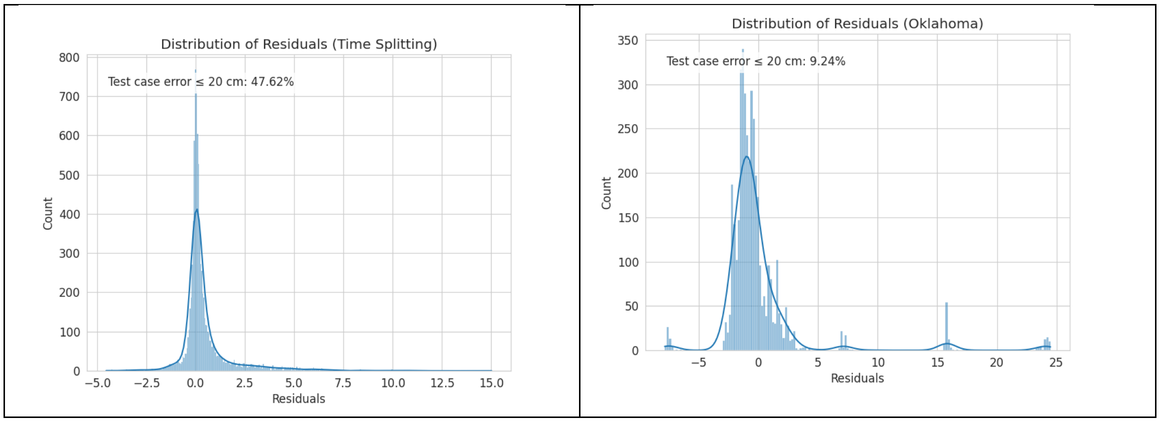
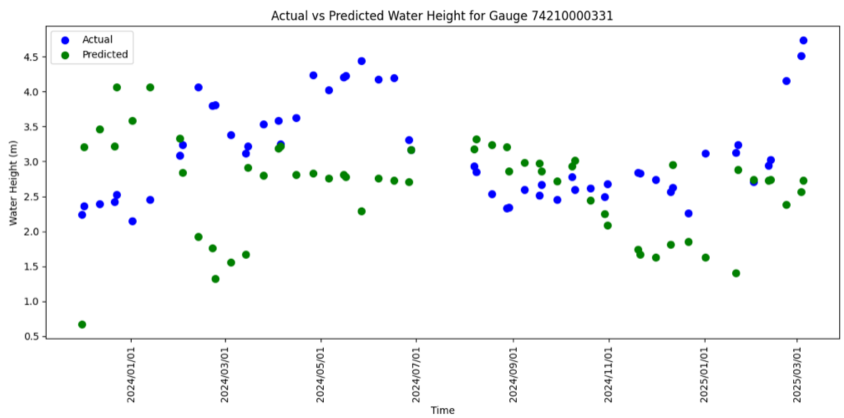
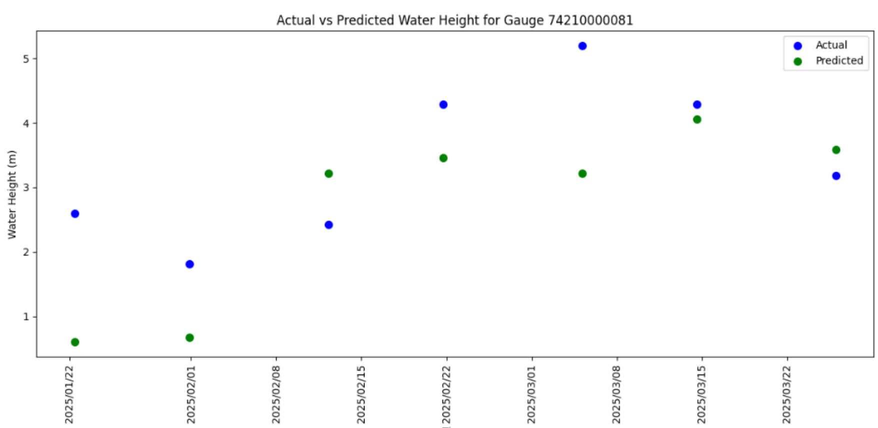


Figure 2 Comparison of different test scenarios residuals on test set, a) random sampling shows best performance b) time-based splitting stands on the second and d) leave one state out is least performer by this metric





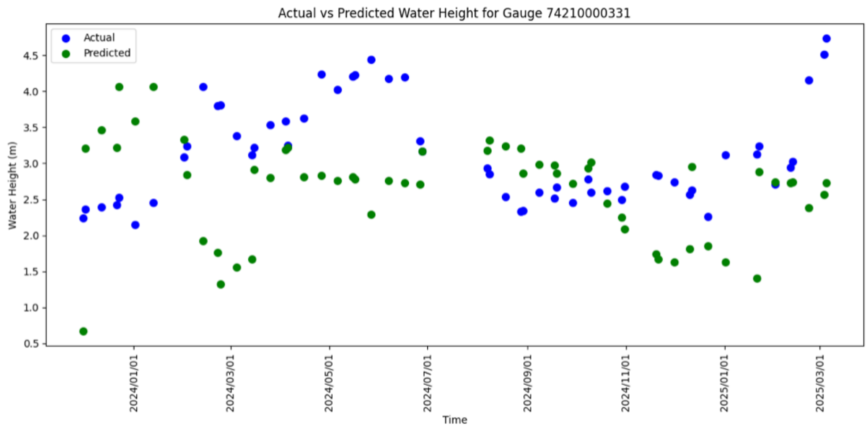


Figure 3- Visual comparison of some random gauge's performance of time spitted and Random Forest is shown here

Figure 3 presents the performance of the Random Forest model on a time-split dataset for a representative gauge. The predicted water heights (green) are plotted against the actual measurements (blue) over time. The model captures the general trend of the water level variations but tends to underestimate peaks and overestimate low points, indicating a bias toward average values. This behavior is typical of ensemble models and suggests moderate underfitting. Despite these deviations, the model maintains consistent predictive performance over the time period, with no clear degradation. Overall, the figure demonstrates that the model generalizes reasonably well to future observations but struggles to capture sharp fluctuations in the time series.

Feature Importance

Analysis of RF and GB importance scores revealed that the geoid information, terrain elevation and derived water depth proxy (SWOT height minus terrain elevation) were among the most informative predictor, followed by raw water surface elevation, river width, and geoid-corrected water surface elevation. Distance-based variables (inverse distance to lakes) and catchment characteristics contributed less, suggesting hydraulic controls dominate depth variability in our dataset.

Discussion:

Comparison to Existing Methods

Our RF model, with MAE ≈ 0.32 m in the random split, exceeds the ~10 % relative RMSE (~0.5–1 m absolute) reported by Tourian et al. (2017) when applied to the Po River . Moreover, Wu et al. (2023) [8] achieved MAE < 0.5 m for coastal bathymetry using Sentinel-2 data , indicating that our approach with SWOT data is competitive even in complex fluvial environments. Unlike statistical inversion methods that require extensive in situ discharge data and Gauss–Helmert frameworks, the ML approach flexibly captures nonlinear interactions among predictors.

Spatial and Temporal Generalization

While random sampling highlights the potential accuracy under idealized conditions, performance degrades in river-based and leave-one-state-out splits, revealing limitations in spatial transferability. This pattern aligns with Elkhrachy (2022) [9], who noted reduced ML model accuracy in ungauged basins when trained on neighboring catchments . Conversely, the strong temporal split results demonstrate that once a channel's characteristics are learned, ML models can effectively track temporal variations, making them well-suited for forecasting short-term depth fluctuations where local training data exist.

Limitations

- **Data Frequency and Alignment:** SWOT's 21-day revisit introduces temporal gaps; aligning with 15 min gauge data required loosened matching (±7.5 min), potentially introducing mismatch errors.
- **Channel Morphology Variability:** Our static predictors (terrain elevation, slope) do not fully capture seasonal or event-driven morphological changes (e.g., sediment deposition).
- **Training Data Coverage:** Limited gauge coverage in Louisiana (35 gauges) may have biased the leave-one-state-out results, suggesting a need for more distributed in situ observations.

Future Work

- **Higher Resolution DEM:** Using more fine scale DEM like FABDEM can help improve the model predictiveness
- **Hybrid Models:** Combine ML with physics-based inversion (e.g., Manning's equation constraints) to enforce hydrodynamic consistency.
- **Spatio-Temporal Architectures:** Explore recurrent or graph-based neural networks to better capture dynamic and network-structured relationships among reaches.

Conclusions

This study demonstrates that machine learning, particularly Random Forests, can accurately estimate river depth from SWOT-derived water surface elevation, width, slope, and terrain predictors, achieving MAE as low as 0.32 m under random sampling. While models generalize well temporally, spatial transferability remains challenging, underscoring the importance of localized training data or hybrid modeling approaches. By integrating high-resolution satellite observations with flexible ML algorithms, our approach offers a scalable alternative to traditional bathymetric inversion, with promising applications in hydrological forecasting, flood management, and remote monitoring of ungauged basins. Future enhancements—such as richer feature sets and physics-informed architectures—are expected to further improve accuracy and generalizability, advancing the utility of SWOT data in water resource science.

Data Dictionary:

Variable	Description
terrain_elevation_m	Terrain elevation at the gauge location, in meters, typically derived from a digital elevation model.
swot_height_m	Water surface elevation observed by the SWOT satellite, in meters, measured relative to a reference ellipsoid (also referred to as WSE).
swot-terrain	Difference between swot_height_m and terrain_elevation_m, representing water depth in meters.
inv_distance	Inverse distance to a reference point (e.g., nearest gauge or feature), in meters, used as a spatial weighting factor.
geoid	Geoid height at the gauge location, in meters, representing the equipotential surface of Earth's gravity field.
lake_depth	Depth of the lake or water body at the gauge location, in meters, if applicable.
terrain_slope_deg	Slope of the terrain at the gauge location, in degrees, derived from elevation data.
lake_area	Surface area of the lake or water body at the gauge location, in square meters, if applicable.
wse_corrected	Corrected water surface elevation from SWOT data, in meters, adjusted for systematic biases or errors.
swot_n_good_nod	Number of good nodes (valid observations) in the SWOT data for the gauge location, indicating data quality.
swot_width_m	River width observed by the SWOT satellite, in meters.
swot_slope2	Alternative slope measurement of the water surface from SWOT data, in meters per meter, possibly a secondary calculation method.
swot_slope	Slope of the water surface observed by the SWOT satellite, in meters per meter, indicating the river's gradient.

Citations:

[1] M. K. Mersel, L. C. Smith, K. M. Andreadis, and M. T. Durand, “Estimation of river depth from remotely sensed hydraulic relationships,” *Water Resources Research*, vol. 49, no. 6, pp. 3165–3179, 2013, doi: 10.1002/wrcr.20176.

[2] M. Durand, L.-L. Fu, D. P. Lettenmaier, D. E. Alsdorf, E. Rodriguez, and D. Esteban-Fernandez, “The Surface Water and Ocean Topography Mission: Observing Terrestrial Surface Water and Oceanic Submesoscale Eddies,” *Proceedings of the IEEE*, vol. 98, no. 5, pp. 766–779, May 2010, doi: 10.1109/JPROC.2010.2043031.

[3] M. J. Tourian, O. Elmi, A. Mohammadnejad, and N. Sneeuw, “Estimating River Depth from SWOT-Type Observables Obtained by Satellite Altimetry and Imagery,” *Water*, vol. 9, no. 10, Art. no. 10, Oct. 2017, doi: 10.3390/w9100753.

[4] M. T. Durand, K. Andreadis, L. C. Smith, D. E. Alsdorf, and M. K. Mersel, “A Bayesian algorithm for estimation of river depth, roughness and discharge from SWOT measurements of height and inundated area,” presented at the AGU Fall Meeting Abstracts, Dec. 2011, pp. H53O-06. Accessed: Apr. 25, 2025. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2011AGUFM.H53O..06D

[5] J. Neal, L. Hawker, J. Savage, M. Durand, P. Bates, and C. Sampson, “Estimating River Channel Bathymetry in Large Scale Flood Inundation Models,” *Water Resources Research*, vol. 57, no. 5, p. e2020WR028301, 2021, doi: 10.1029/2020WR028301.

[6] G. Choné *et al.*, “Large-scale flood modelling based on LiDAR data: a case study in the Southwest Miramichi watershed, New Brunswick, Canada,” *Canadian Water Resources Journal / Revue canadienne des ressources hydriques*, vol. 49, no. 4, pp. 460–478, Oct. 2024, doi: 10.1080/07011784.2024.2430776.

[7] J. C. Neal *et al.*, “Flood Inundation Modelling using the Surface Water Ocean Topography Mission”, Accessed: Apr. 29, 2025. [Online]. Available: https://www.authorea.com/doi/full/10.22541/essoar.174319735.54049202/v1?commit=9058d0c7bc2ef6eba2809db868d007546400547c

[8] Z. Wu, S. Wu, H. Yang, Z. Mao, and W. Shen, “Enhancing Water Depth Estimation from Satellite Images Using Online Machine Learning: A Case Study Using Baidu Easy-DL with Acoustic Bathymetry and Sentinel-2 Data,” *Remote Sensing*, vol. 15, no. 20, Art. no. 20, Jan. 2023, doi: 10.3390/rs15204955.

[9] I. Elkhrachy, “Flash Flood Water Depth Estimation Using SAR Images, Digital Elevation Models, and Machine Learning Algorithms,” *Remote Sensing*, vol. 14, no. 3, Art. no. 3, Jan. 2022, doi: 10.3390/rs14030440.

[10] “WISP.” Accessed: Apr. 29, 2025. [Online]. Available: https://apps.usgs.gov/wisp?lat=37.345689&lon=-109.705836&zoom=2.67&basemap=%22Light%22&useDevStyles=false&activeLayers=%5B%7B%22id%22%3A%22sword%22%2C%22value%22%3A%22width%22%7D%5D

[11] D. Yamazaki *et al.*, “A high-accuracy map of global terrain elevations,” *Geophysical Research Letters*, vol. 44, no. 11, pp. 5844–5853, Jun. 2017, doi: 10.1002/2017GL072874.

[12] N. US Department of Commerce and N. US Department of Commerce, "GEOID18 | GEOID | Data & Imagery | National Geodetic Survey." Accessed: Apr. 29, 2025. [Online]. Available: https://geodesy.noaa.gov/GEOID/GEOID18/downloads.shtml

[13] "HydroLAKES." Accessed: Apr. 29, 2025. [Online]. Available: https://www.hydrosheds.org/products/hydrolakes

[14] A. T. Zhang and V. X. Gu, "Global Dam Tracker: A database of more than 35,000 dams with location, catchment, and attribute information," *Sci Data*, vol. 10, no. 1, p. 111, Feb. 2023, doi: 10.1038/s41597-023-02008-2.